

Proof of Concept Evaluation and
Redesign
Personal Project - Minor AI for Society

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1. Introduction

This document describes the findings, limitations, and redesign decisions resulting from the initial Proof of Concept (PoC) for the AI-based plant monitoring system.

The original goal was to predict plant health using machine learning models based on environmental sensor data.

2. PoC Results

The PoC implemented several machine learning models, including:

- Logistic Regression
- Multi-Layer Perceptron (MLP)
- Random Forest

The models were trained on a plant health dataset containing:

- Soil moisture
- Temperature
- Humidity
- Light intensity
- Derived stress features

3. Key Findings

3.1. Strong Dependence on Soil Moisture

Correlation analysis revealed:

- Soil moisture and moisture-related features had extremely high correlation with the target
- Other features (temperature, humidity, light) showed minimal influence

This resulted in:

- Models primarily learning soil moisture thresholds
- Limited contribution from other environmental variables

3.2. Feature Redundancy

Several engineered features were highly correlated:

- Soil moisture vs moisture stress

- Soil moisture vs interaction features

This reduced the effective dimensionality of the dataset.

3.3. Dataset Limitations

The dataset exhibited characteristics of:

- Synthetic or simplified labeling
- Single-factor dependency (soil moisture)
- Lack of real-world complexity

As a result:

- Machine learning models replicated simple threshold behavior
- No meaningful multi-factor learning occurred

3.4. Lack of Temporal Information

The dataset did not include timestamps, preventing:

- Trend analysis
- Duration-based stress detection
- Recovery tracking after watering

This limited the system to static snapshot predictions.

4. Problem Statement

The core issue identified is:

The dataset does not accurately represent real-world plant health, leading to models that do not generalize beyond simple soil moisture thresholds.

5. Redesign Strategy

To address these limitations, the system is redesigned with a hybrid approach combining:

- Rule-based classification
- Temporal analysis
- Machine learning for pattern recognition

5.1. Multi-Factor Classification

Instead of a single health label, the system now classifies:

- Temperature (LOW / OK / HIGH)
- Humidity (LOW / OK / HIGH)
- Soil moisture (LOW / OK / HIGH)

This provides a more interpretable and modular representation of plant conditions.

5.2. Stress-Based Modeling

Each factor is evaluated using a stress function:

Stress = $|\text{value} - \text{ideal}|$

This allows:

- Quantification of deviation
- Plant-specific customization
- Consistent scaling across features

5.3. Temporal Analysis

A history buffer is introduced to track:

- Rolling averages
- Value changes (delta)
- Trends (slope)
- Duration of stress conditions

This enables detection of:

- Drying patterns
- Recovery after watering
- Sustained unfavorable conditions

5.4. Machine Learning Role

Machine learning is repositioned to handle:

- Pattern recognition over time
- Anomaly detection
- Risk prediction

- Trend classification

Rather than replacing rule-based logic, ML enhances the system's ability to interpret temporal behavior.

5.5. Recommendation Engine

A recommendation layer translates system output into actionable advice:

- Watering suggestions
- Environmental adjustments
- Risk warnings

6. Next Steps

The following steps will be implemented:

1. Develop modular C++ architecture:
 - Sensor layer
 - Classification layer
 - Temporal buffer
 - Feature engineering
 - ML inference module
2. Implement temporal feature extraction on-device
3. Train ML models on:
 - temporal features
 - multi-factor stress indicators
4. Compare rule-based vs ML-enhanced performance
5. Evaluate system using real sensor data

7. Conclusion

The initial PoC demonstrated that the dataset was insufficient for multi-factor plant health prediction.

The redesigned system:

- Removes dependence on flawed dataset assumptions
- Incorporates real-world temporal behavior
- Maintains machine learning as a core component

This results in a more robust, explainable, and scalable plant monitoring system.